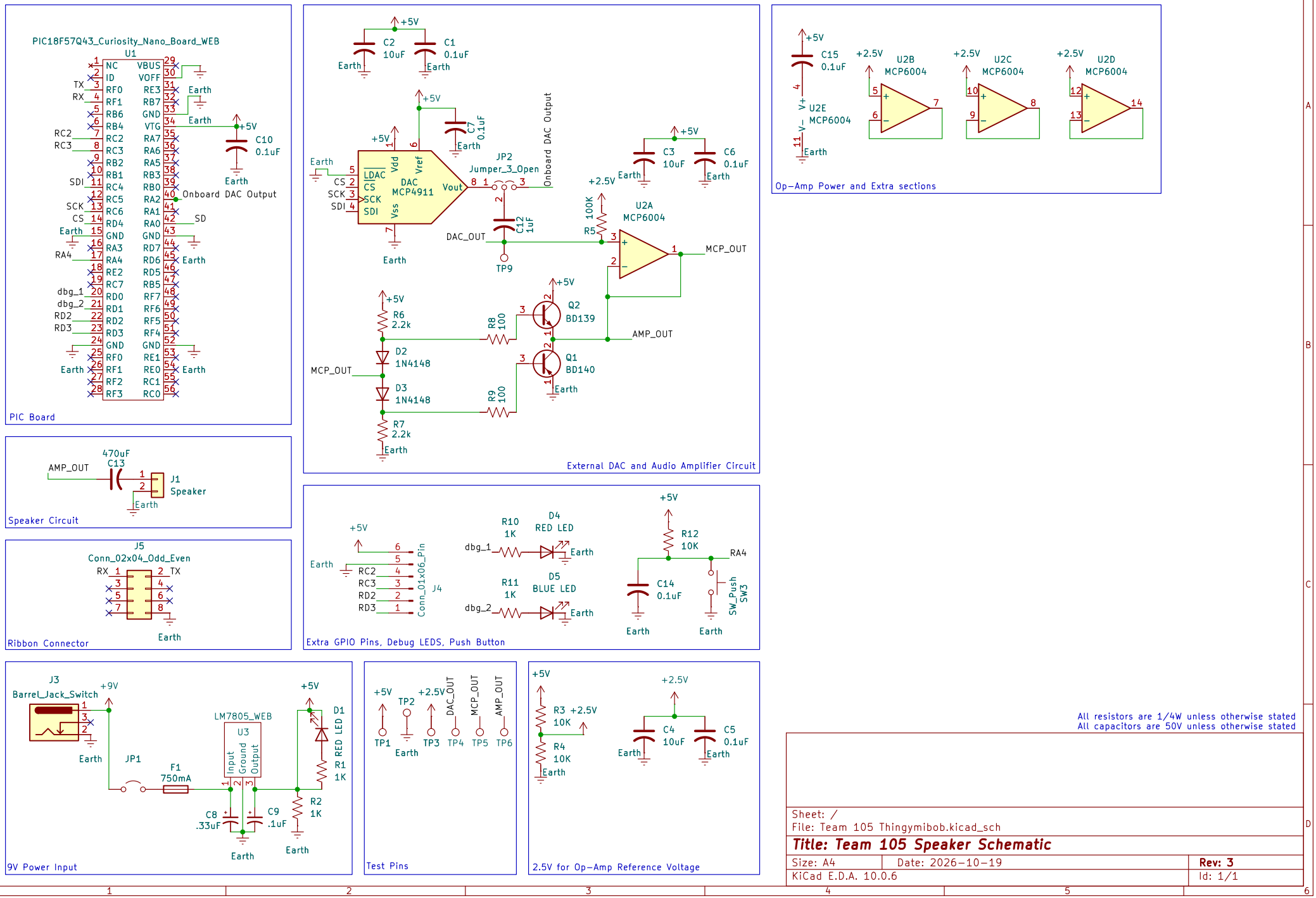

PIC18F57Q43_Curiosity_Nano_Board_WEB
U1
PIC Board
+5V
C2 10uF
C1 0.1uF
Earth
C7 0.1uF
JP2 Jumper_3_Open
Onboard DAC Output
DAC MCP4911
LDAC
CS
SCK
SDI
Vdd
Vref
Vout
Vss
C12 1uF
DAC_OUT
TP9
C3 10uF
C6 0.1uF
+2.5V
U2A MCP6004
R5 100K
MCP_OUT
R6 2.2k
R8 100
Q2 BD139
AMP_OUT
D2 1N4148
D3 1N4148
Q1 BD140
R9 100
R7 2.2k
External DAC and Audio Amplifier Circuit
C15 0.1uF
U2E MCP6004
U2B MCP6004
U2C MCP6004
U2D MCP6004
Op-Amp Power and Extra sections
470uF C13
J1 Speaker
Speaker Circuit
J5 Conn_02x04_Odd_Even
Ribbon Connector
J4 Conn_01x06_Pin
R10 1K
D4 RED LED
R11 1K
D5 BLUE LED
R12 10K
C14 0.1uF
SW3 SW_Push
Extra GPIO Pins, Debug LEDS, Push Button
J3 Barrel_Jack_Switch
+9V
JP1
F1 750mA
LM7805_WEB
U3
Input
Ground
Output
C8 .33uF
C9 .1uF
R2 1K
D1 RED LED
R1 1K
9V Power Input
TP1
TP2
TP3
TP4
TP5
TP6
Test Pins
R3 10K
R4 10K
C4 10uF
C5 0.1uF
2.5V for Op-Amp Reference Voltage
All resistors are 1/4W unless otherwise stated
All capacitors are 50V unless otherwise stated
Sheet: /
File: Team 105 Thingymibob.kicad_sch
Title: Team 105 Speaker Schematic
Size: A4
Date: 2026-10-19
Rev: 3
KiCad E.D.A. 10.0.6
Id: 1/1